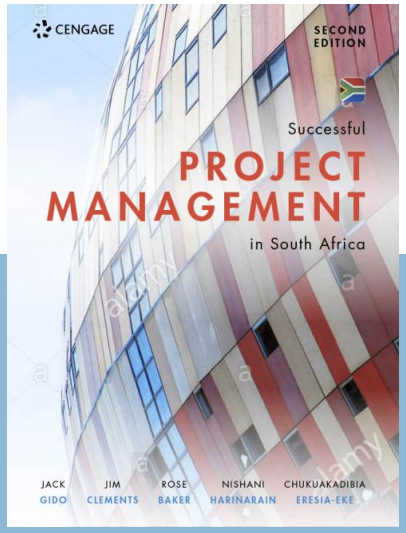


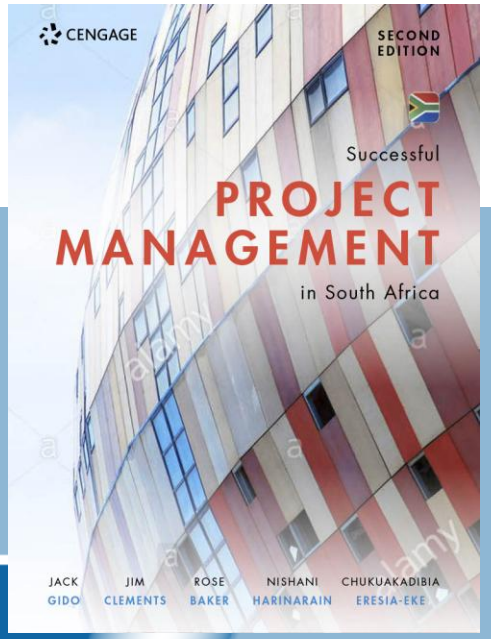
Learning Unit 6

Developing the Schedule and Project Utilisation

(Chapter 5 and 6)

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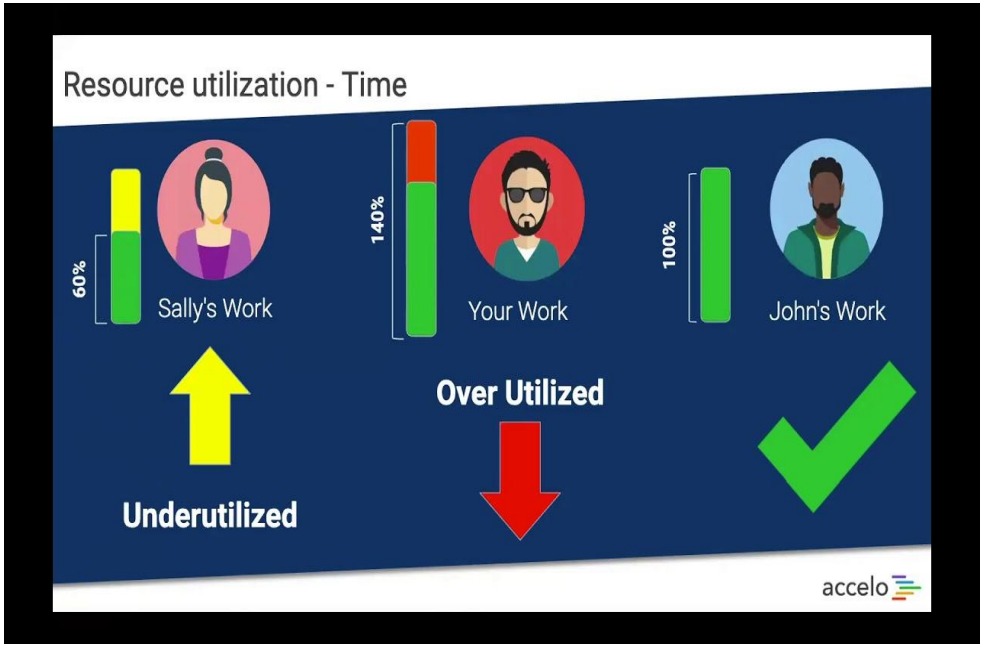
CHAPTER 6



Resource Utilisation

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- The consideration of resources adds another dimension to planning and scheduling
- It is necessary to estimate the types and quantities of resources required to perform each activity
- These resources can include people, materials, equipment, tools, facilities and so forth



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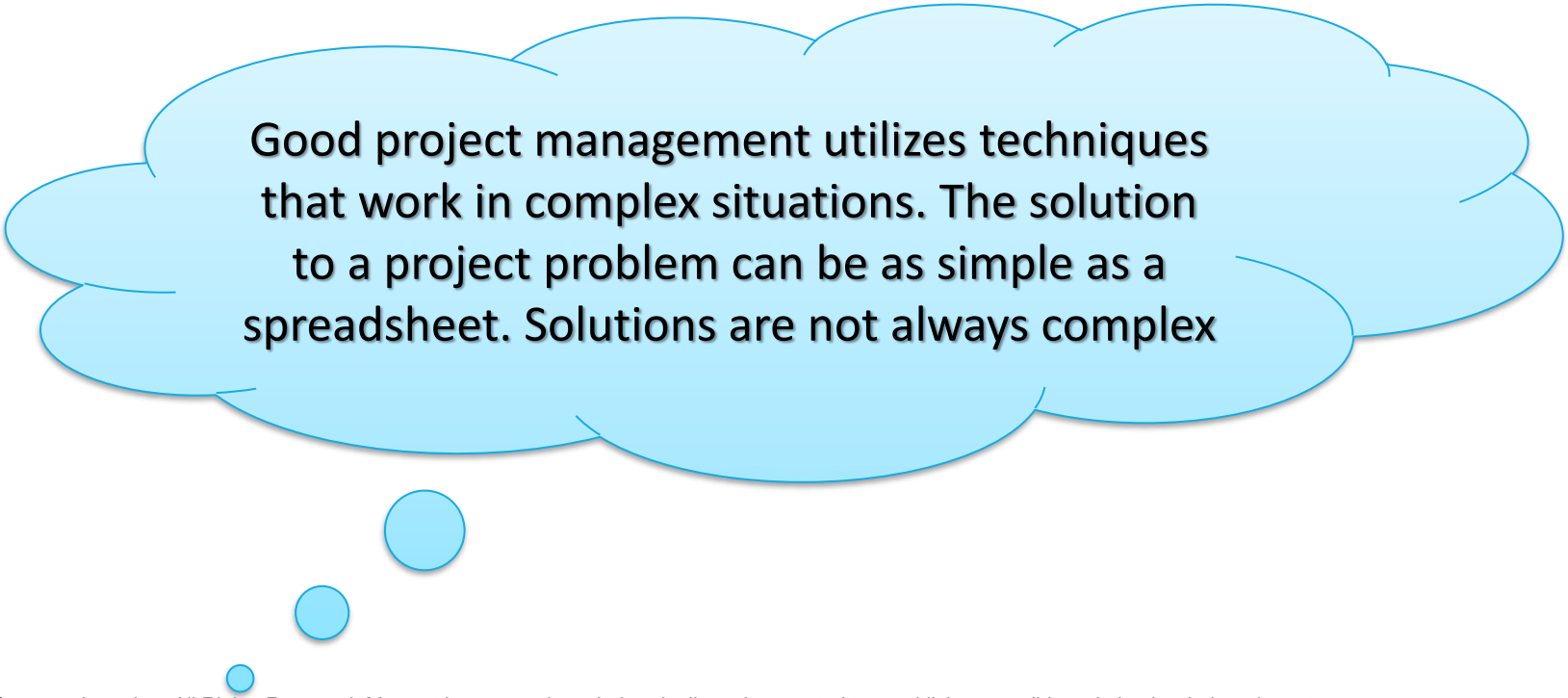
1. Why is it important to consider resources when planning and managing a project?
2. Will the number of resources available affect the completion date?
3. If you had 2 experienced people building a house, would it take as long as if you had 10 experienced people building a house?

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- A resource requirements plan illustrates the expected utilization of resources by time period during the time span of the project
- In many projects, the quantities of the various types of resources available to perform the project activities are limited
- Several activities may require the same resources at the same time and there may not be sufficient resources available to satisfy all the demands
- These activities are competing for the use of the same resources
- If insufficient resources are not available when required, some activities may have to be delayed until a later time when resources are available for them

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- Therefore, resources can constrain the project schedule
- They can also be a constraint to completing the project within budget if it is determined that additional resources are needed to complete the project by its required completion time

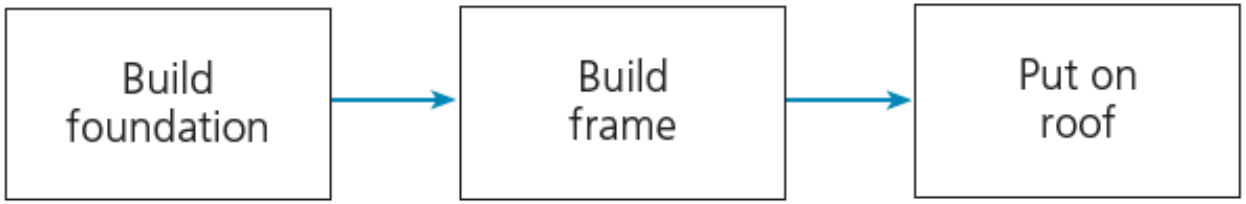


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Technically Constrained Planning

- Serial relationship – performed in that sequence

FIGURE 6.1 Technically Constrained Activity Sequence

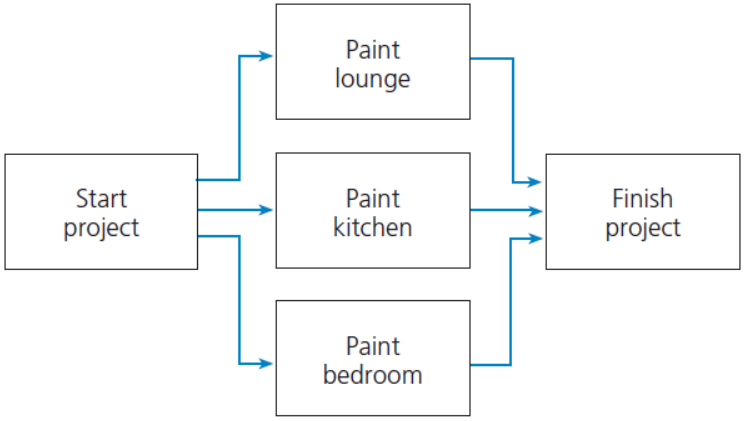


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Resource-Constrained Planning

- Available resources allow for concurrent tasks

FIGURE 6.2 Resource-Constrained Planning



(a) Activity sequence without resource constraints



(b) Activity sequence based on resource constraints

- Limited resource availability constrains project

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Resource Requirements Plan

- Illustrates the expected resource use by time period
- Indicate the amounts and types of resources needed to perform each activity

FIGURE 6.3 Estimated Resource Requirements for Consumer Market Study Project

NAME	ACTIVITIES	WORK DAYS	PERIOD
Mbali	1, 2, 3, 4, 8	40	0 to 40
Fani	5, 6, 9	8	38 to 53
S'bu	7, 10	17	38 to 55
Jim	11, 12, 13	25	103 to 128
		90	**0 to 128**

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Example painting project: resource requirements

FIGURE 6.4 Required Resources for Painting Project

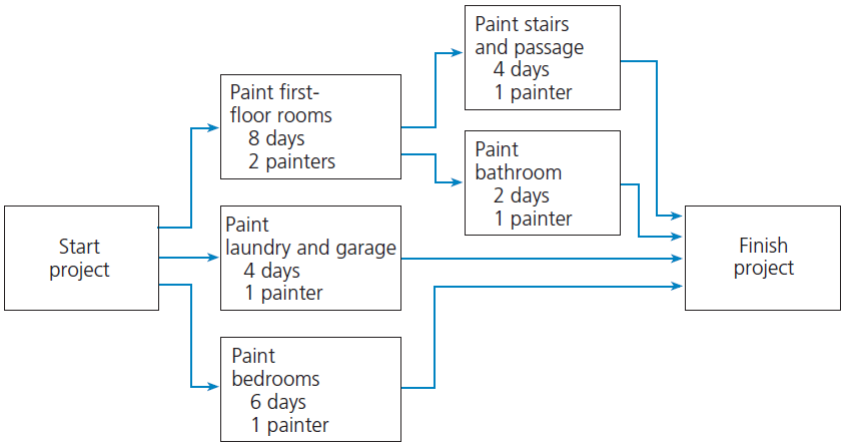
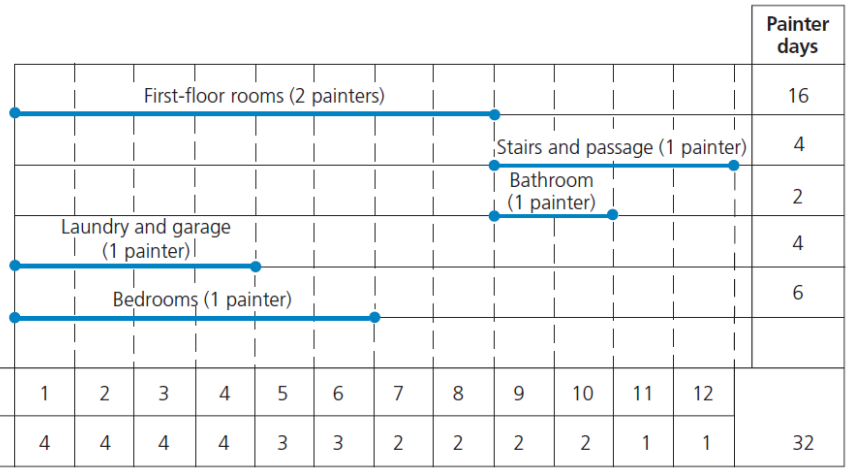
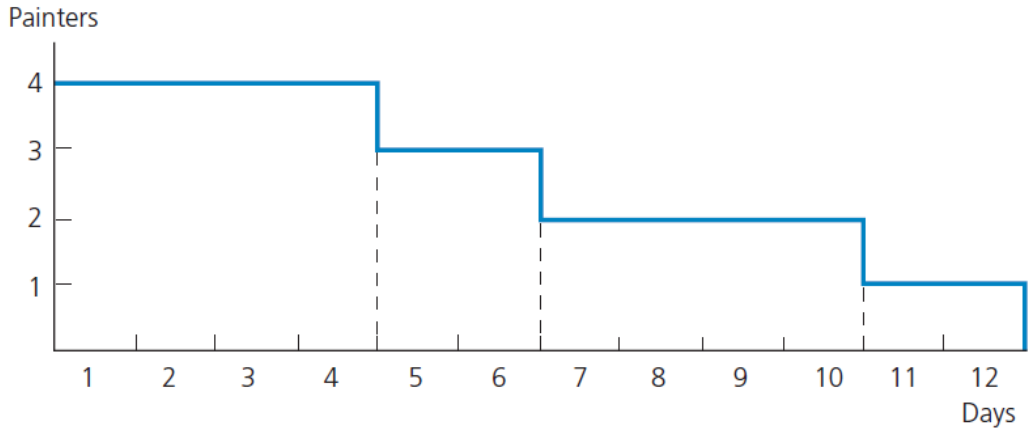


FIGURE 6.5 Resource Requirements Plan for Painting Project



Day	1	2	3	4	5	6	7	8	9	10	11	12	
Painters	4	4	4	4	3	3	2	2	2	2	1	1	32

FIGURE 6.6 Resource Utilisation for Painters



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Resource Levelling

- Minimise resource requirement fluctuations
- Resources applied as uniformly as possible
- Attempt to keep project schedule within required time
- Delay start of noncritical activities
- Use positive slack
- Project management information systems assist
- If delay beyond slack and on critical path, project will be delayed beyond required completion time

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Resource-Limited Scheduling

- Develop shortest schedule
- Not exceed fixed available resources
- Extend the project completion time if necessary
- Give activities with the least slack first priority
- Delay lower priority activities

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Resource Requirements for Information Systems Development

- Five basic required resources
  - People, hardware, software, data, and network resources
- The more accurate the resource assessment, the more likely the project can be completed on time
- Most common problem – over allocation
  - Resources assigned to multiple tasks at same time
  - Conflict results in lengthening the time

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Project Management Information Systems

- Handle resource considerations within a project
- Create and maintain a list of resources
- Store availability, rate, and costs for resources
- Assign resources and calculate cost
- Assign calendar to record availability
- Resolve overallocations for best solution

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Critical Success Factors

- **Resources can constrain the project schedule** because the quantities of various types of resources available to perform the project activities may be limited.
- It is necessary **to estimate the types and quantities** of resources required to perform each activity.
- If sufficient resources are not available when required, some activities **may have to be delayed** until a later time when resources become available to perform the activities.
- **Resource levelling**, or smoothing, is a method for developing a schedule that attempts to minimise the fluctuations in requirements for resources. It levels the resources so that they are **applied as uniformly as possible** without extending the project schedule beyond its required completion time.
- **Resource-limited scheduling** is a method for developing the shortest schedule when the quantity of resources is fixed. It **will extend the project** completion time if necessary.

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Summary

- The consideration of resources adds another dimension to planning and scheduling; resources can constrain a project.
- Resources can include people, materials, equipment, tools, facilities, and other crucial elements to a project.
- A resource requirements plan illustrates the expected utilisation of resources by time period during the time span of the project.
- Resource levelling, or smoothing, is a method for developing a schedule that attempts to minimise the fluctuations in requirements for resources when the project completion time is fixed.
- Resource-limited scheduling is a method for developing the shortest schedule when the quantity of available resources is fixed, and may extend the project completion time in order to keep within the resource limits.

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